

AI LeadOps Engine

End-to-End Lead Generation, Outreach, CRM,
and Pipeline Automation System

Case Study

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June 2026

Built with n8n, Google Maps API, Google Sheets, Gmail, and JavaScript

1 Executive Summary

Small businesses often spend significant time finding prospects, managing spreadsheets, conducting outreach, and tracking follow-ups manually. These repetitive tasks create operational inefficiencies and increase the risk of missed opportunities.

The AI LeadOps Engine is an automation system designed to streamline lead generation and sales operations. Built using n8n, Google Maps API, Google Sheets, Gmail, and JavaScript, the system automates lead discovery, data enrichment, lead scoring, outreach, CRM updates, pipeline tracking, and reporting.

The project demonstrates how workflow automation can turn scattered manual sales tasks into a structured operational system. Instead of manually searching for leads, copying data into spreadsheets, writing outreach messages, and tracking follow-ups, the system connects these steps into one automated process.

2 Problem Statement

Many small and medium-sized businesses rely on manual processes for lead generation and follow-up. This often creates several problems:

- Leads are found manually through online searches.
- Contact information is copied into spreadsheets by hand.
- Outreach messages are written manually.
- Follow-ups are forgotten or delayed.
- Pipeline status is not clearly tracked.
- Sales activity is difficult to measure.

These issues reduce speed, consistency, and visibility across the sales process. A business may have potential leads available, but without a reliable system to capture, contact, and track them, many opportunities can be lost.

3 Solution Overview

The AI LeadOps Engine solves this by connecting lead generation, outreach, CRM management, and pipeline tracking into one automated system.

The system is composed of three core workflows:

Workflow	Purpose
Workflow 1: WhatsApp Lead Intake	Captures inbound lead messages, classifies the inquiry, stores the lead in Google Sheets, and sends an email notification.
Workflow 2: Google Maps Lead Generator	Discovers businesses from Google Maps, enriches the lead data, scores the leads, and stores them in the CRM.
Workflow 3: AI Follow-Up Pipeline	Generates personalized outreach, sends email notifications, updates CRM status, schedules follow-ups, and tracks pipeline stages.

Together, these workflows form a lightweight LeadOps system that can support prospecting, outreach, and follow-up management.

4 Technology Stack

Component	Technology
Automation Engine	n8n
Lead Source	Google Maps API
CRM	Google Sheets
Email Outreach	Gmail
Business Logic	JavaScript
Reporting	Google Sheets Dashboard

5 System Architecture

The system uses n8n as the automation engine. Google Maps provides business discovery data, Google Sheets acts as the CRM and pipeline database, and Gmail handles outreach notifications.

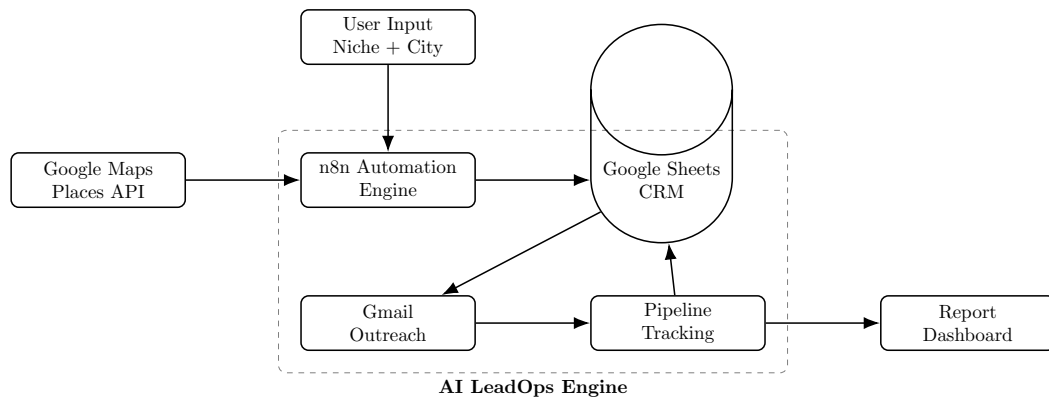


Figure 1: AI LeadOps Engine system architecture

6 Workflow Design

6.1 Workflow 1: WhatsApp Lead Intake

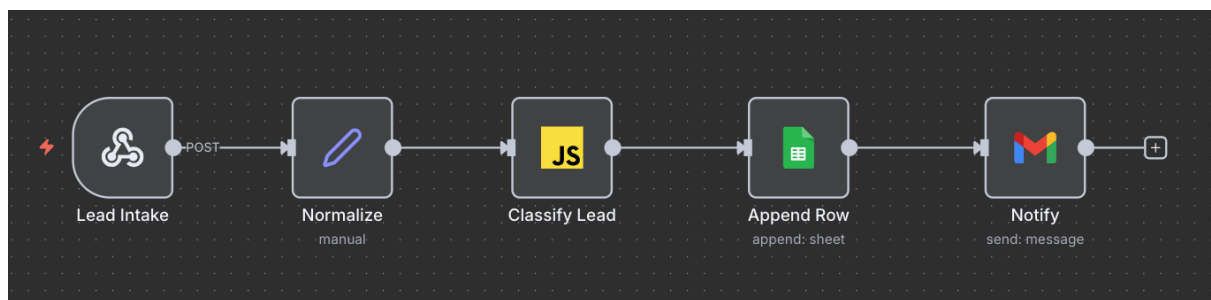


Figure 2: WhatsApp lead intake workflow

The first workflow captures incoming lead information through a webhook. The incoming message is normalized, classified, stored in Google Sheets, and sent as an email notification.

- Captures inbound lead data.
- Extracts name, phone number, message, date, and source.
- Classifies the inquiry as pricing, booking, or general.
- Stores the lead in Google Sheets.
- Sends an email notification.

6.2 Workflow 2: Google Maps Lead Generator

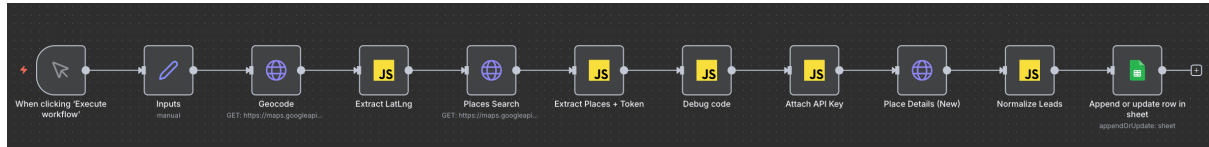


Figure 3: Google Maps lead generation workflow

The second workflow discovers outbound leads using Google Maps. It searches for businesses by niche and location, extracts business details, enriches the data, scores each lead, and stores the results in the CRM.

- Searches for businesses using Google Maps.
- Extracts business name, address, phone, website, and operating status.
- Normalizes phone numbers and opening hours.
- Scores leads using rule-based logic.
- Prevents duplicates using phone number matching.

6.3 Workflow 3: AI Follow-Up Pipeline

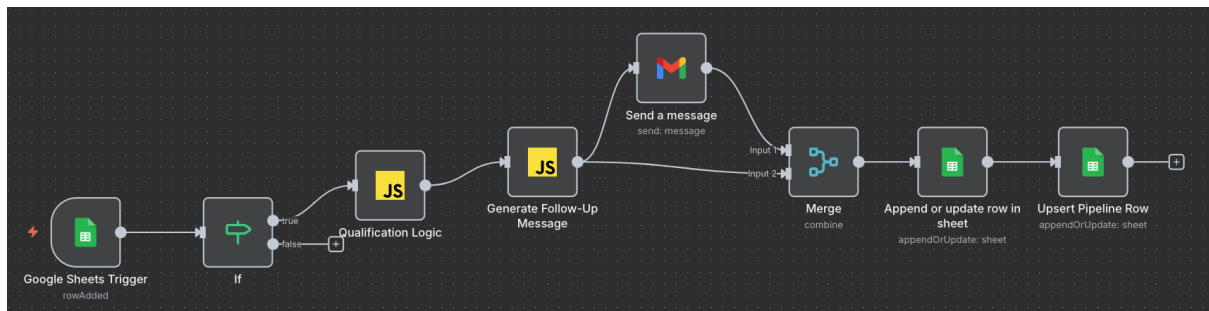


Figure 4: Lead follow-up and pipeline workflow

The third workflow monitors new leads inside the CRM. When a new uncontacted lead is detected, the system generates a personalized outreach message, sends an email, updates the lead status, and creates or updates a pipeline record.

- Detects new uncontacted leads.
- Assigns lead priority.
- Generates personalized outreach messages.
- Sends email outreach.
- Updates lead status to contacted.
- Tracks pipeline stage and follow-up dates.

7 CRM and Pipeline Design

The system uses Google Sheets as a lightweight CRM. The CRM stores lead information, business details, lead scores, outreach status, follow-up attempts, and lifecycle stages.

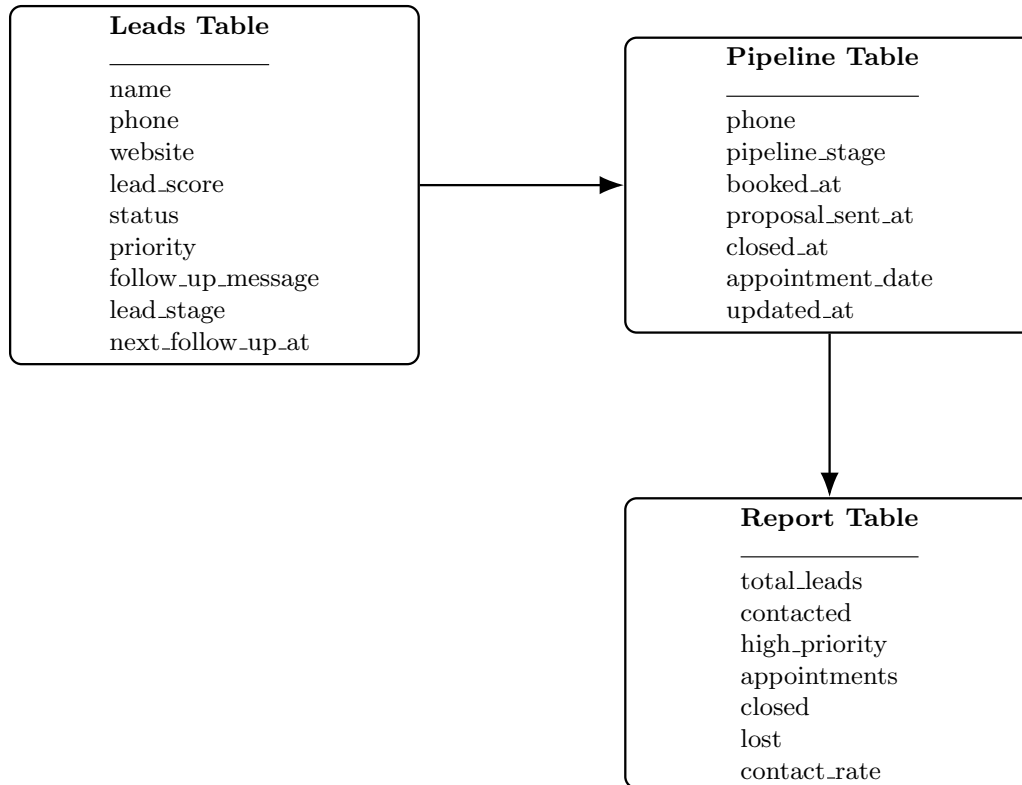


Figure 5: CRM schema and data model

The pipeline tracks the sales lifecycle from new lead to contacted, waiting reply, follow-up, appointment booking, proposal, closed, or lost.

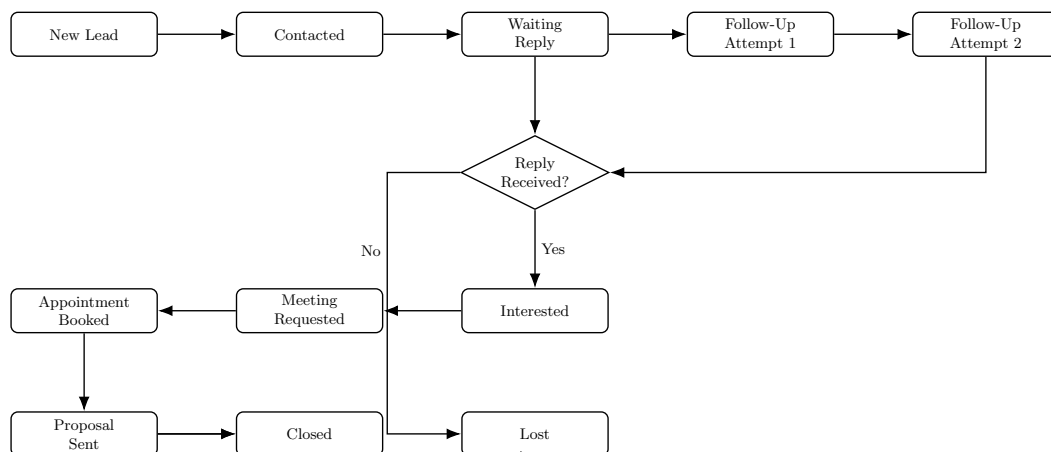


Figure 6: Lead lifecycle and pipeline stages

8 Results and Demonstration

The completed system successfully demonstrates an end-to-end lead operations process.

8.1 Project Outcomes

The system was tested using physiotherapy clinics in the UAE as a target niche. During testing, the workflow successfully discovered business leads through Google Maps, enriched lead information, generated outreach messages, and created pipeline records for follow-up tracking. A total of 20 business leads were processed through the automation pipeline, demonstrating the ability to automate lead discovery, outreach preparation, CRM updates, and pipeline management without manual data entry.

- Businesses can be discovered automatically from Google Maps.
- Lead data is enriched and stored in a CRM.
- Leads are scored and prioritized.
- Personalized outreach messages are generated.
- Email outreach is sent automatically.
- Lead status is updated after outreach.
- Pipeline records are created and updated.
- Reporting fields provide visibility into activity.

Dr Sunny Medical	97165668966	Sheikh Zayed St	https://www.dr.su	OPERATIONAL	Google Maps					2026-01-26T15:20:00.358Z			
مركز الزين الطبي	97165667050	Halwan Suburb	https://alwaznalr	OPERATIONAL	Google Maps					2026-01-26T15:20:00.358Z			
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The Garden Mex	97165633777	- Villa 2, block A	https://thegardex	OPERATIONAL	Google Maps					2026-01-26T15:20:00.361Z			
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Al Najma Dental	971525178999	Al Zahra Street	http://www.najm	OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	4
Sama Al Sham I	97167673474	Al Mahatta - Hay	https://samaalsh	OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	4
er Dental Center	971527363399	Tech Tower - Ground Floor - Al R		OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	3
Al Jawhara Dent	971523248525	besides Tasheel - suite 308 ,Jaw		OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	3
NMC Dental Cer	97180052	Formerly known	https://nmc.ae/e	OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	4
h medical center	971581781030	S116 - Muwailah	https://basmaht	OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	4
Crystal Dental C	971564375113	344 Corniche St	http://www.crysst	OPERATIONAL	Google Maps					2026-05-19T16:16:00.000Z	Sharjah	UAE	4

Figure 7: Business leads automatically populated into the CRM

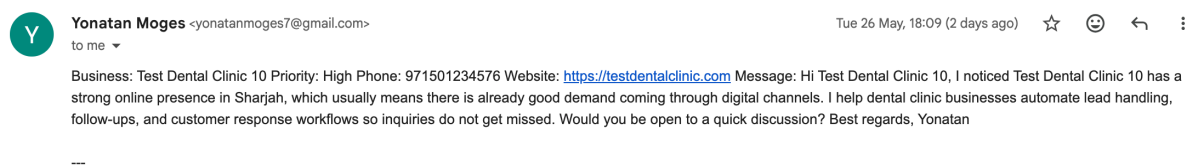


Figure 8: Automated outreach emails generated by the system

	A	B	C	D	E	F	G	H	I	J
1	phone	pipeline_stage	booked_at	proposal_sent_at	closed_at	lost_reason	appointment_date	appointment_notes	updated_at	
2										
3	971501234576	contacted							2026-05-26T15:09:48.696Z	

Figure 9: Pipeline stage tracking and follow-up management

Demo videos are available in the GitHub repository README.

9 Business Value

The AI LeadOps Engine provides value by reducing repetitive manual work and improving sales process consistency.

- Reduces manual lead research.
- Improves lead tracking consistency.
- Speeds up outreach.
- Prevents missed follow-ups.
- Creates a structured sales pipeline.
- Provides visibility into lead activity.

For a business, this type of system can reduce administrative workload and allow teams to focus more on conversations, appointments, and closing opportunities.

10 Future Improvements

Future versions of the system could include:

- AI-generated personalization using business reviews and website content.
- WhatsApp outreach integration.
- Appointment booking automation.
- A client-facing dashboard.
- Database migration from Google Sheets to a production database.
- Real-time analytics and conversion tracking.
- Multi-client support for agency deployment.

11 Conclusion

The AI LeadOps Engine demonstrates how automation can combine lead discovery, CRM management, outreach, pipeline tracking, and reporting into one connected system. The project shows how business process automation can reduce manual work, improve consistency, and create better visibility across the sales workflow.

This project demonstrates how a small business can automate lead generation, outreach, CRM updates, and pipeline tracking using modern automation tools. The architecture can be adapted to industries such as real estate, healthcare, legal services, education, and consulting, providing a foundation for scalable client acquisition systems.